

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 5 Outline

- ▶ **IGP & EGP**
 - Autonomous System (AS)
 - IGP vs EGP
- ▶ **Border Gateway Protocol (BGP)**
 - Introduction
 - BGP and Internet
 - BGP security concerns
 - Class exercise 11
- ▶ **Router ACLs**
 - Operations
 - Application
 - Cisco Syntax
 - Class exercise 12

Interior Gateway Protocols (IGP) & Exterior Gateway Protocols (EGP)

	Interior Gateway Protocols				Exterior Gateway Protocols
	Distance Vector Routing Protocols		Link State Routing Protocols		Path Vector
Classful	RIP	IGRP			EGP
Classless	RIPv2	EIGRP	OSPFv2	IS-IS	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGPv4 for IPv6



Legacy

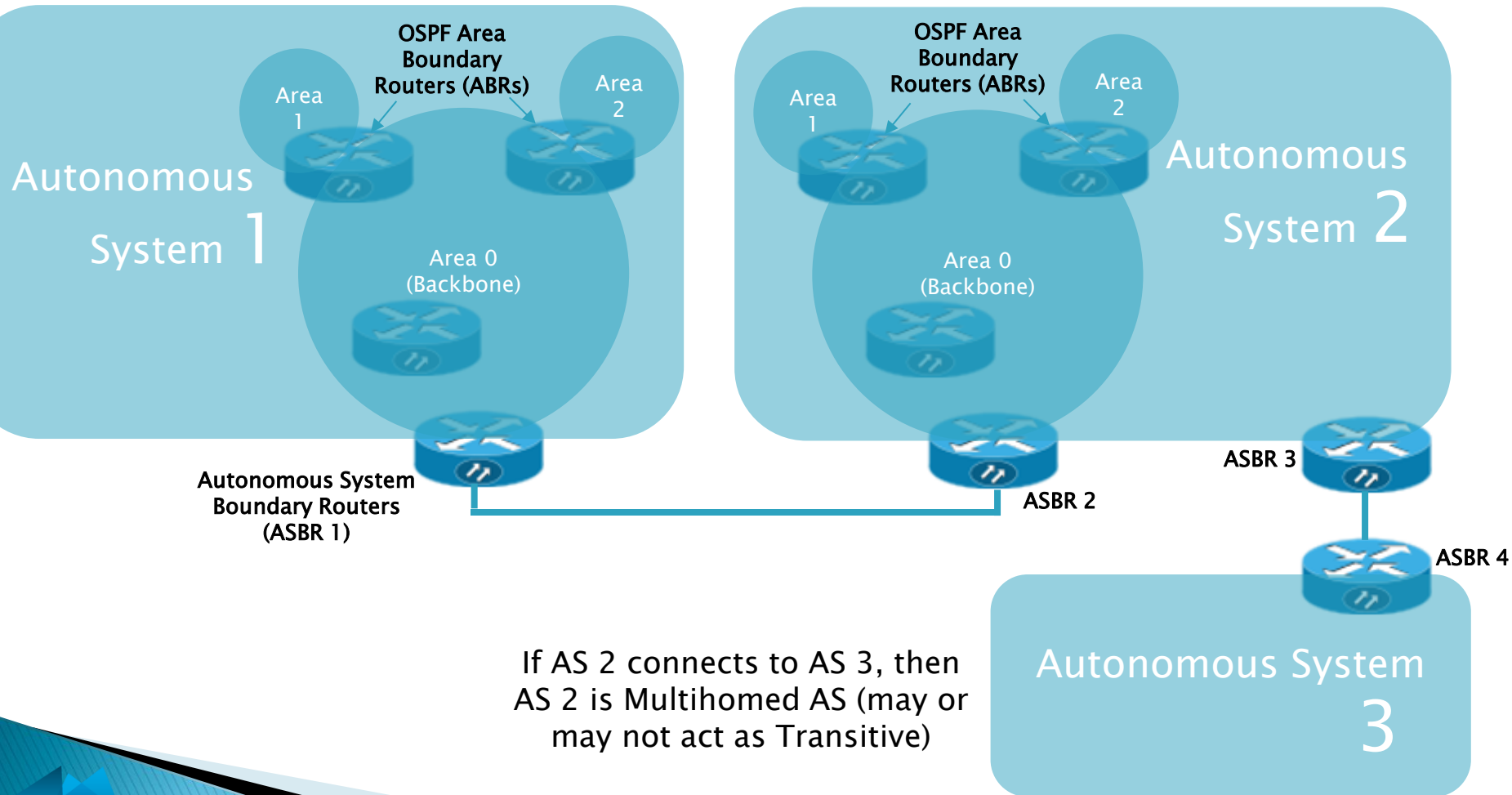
- ▶ Interior Gateway Protocol (IGP) is used within a **single** autonomous system (AS)
 - ▶ They are designed for routing within Autonomous systems
- ▶ Exterior Gateway Protocol (EGP) is used **between different** autonomous systems
 - ▶ Handle routing on the internet level, between large-scale networks

Autonomous System (AS)

- ▶ A network under the control of one or more network operators on behalf of a single administrative entity (e.g., ISP, ...)
 - Presents a common, clearly defined policy to the internet (as defined in RFC 1771, RFC 1930)
 - Every AS is assigned with a unique 16/32 bit AS number by IANA
- ▶ Types:
 - Multihomed: maintains connection to more than one
 - Stub: connected to only one AS (single-homed)
 - Transit: provides connections through itself to other AS's

Autonomous System (AS) (cont.)

If AS 1 is not connected to any other AS than AS 2, then AS 1 is Single-homed (or a Stub)



Autonomous System Number (ASN) assignment

To obtain ASN, from Regional Internet Registry (RIR) delegated by IANA, entity must meet the eligibility criteria set by the RIR

The screenshot displays the RIPE NCC (RIPE Network Coordination Centre) website. The header includes the RIPE NCC logo and navigation links for 'RIPE Database (Whois)' and 'Website'. A search bar is present for 'Search IP Address or ASN'. The main navigation bar lists 'Manage IPs and ASNs', 'Analyse', 'Participate', 'Get Support', 'Publications', and 'About Us'. The breadcrumb trail indicates the user is at 'Home > Manage IPs and ASNs > AS Numbers > Request an AS Number'. The left sidebar contains a menu with 'Manage IPs and ASNs', 'IPv4', 'IPv6', 'AS Numbers' (selected), 'Request an AS Number', 'Reassigning Referenced AS Numbers', 'AS Number Clean up', 'RIPE Database', 'DNS', 'LIR Portal', 'Documentation for Resource Management', 'Resource Transfers and Mergers', 'Legacy Internet Resources', and 'Contact Registry Services'. The main content area is titled 'Autonomous System Numbers' and contains the following text: 'We distribute ASNs according to policies by the RIPE community. Any individual or organisation can request ASNs, though a contractual agreement with the RIPE NCC will be required.' It then states: 'To request an AS Number, you will need to have a contractual agreement with the RIPE NCC. To do this you will either need to become a member or find a sponsoring LIR who will submit the request on your behalf.' A 'Request AS Number' button is shown with the note '(authentication required)'. Below this, it says: 'Here is the information that will be required for your request.' and 'In case you represent a lot of members and you would like to automate the request process, you can use the Resource Request API.' The section 'Request Evaluation' follows, stating: 'We will evaluate your request according to the applicable RIPE policies and RIPE NCC procedures. We will also check against the EU sanctions list – if you or your organisation is found to be under sanctions, the resources request will not be approved.' The footer includes social media icons and a list of links: 'Home', 'Sitemap', 'Contact Us', 'Service Announcements', 'Privacy Statement', 'Legal', 'Cookies', 'Copyright Statement', and 'Terms of Service'. The page is dated 'Created: 23 Dec 2010 - Last updated: 07 Apr 2021' and references 'AS NUMBERS, RESOURCE MANAGEMENT'.

Screenshot from: RIPE NCC, Request an AS Number. Retrieved from <https://www.ripe.net/manage-ips-and-asns/as-numbers/request-an-as-number>

IGP and EGP

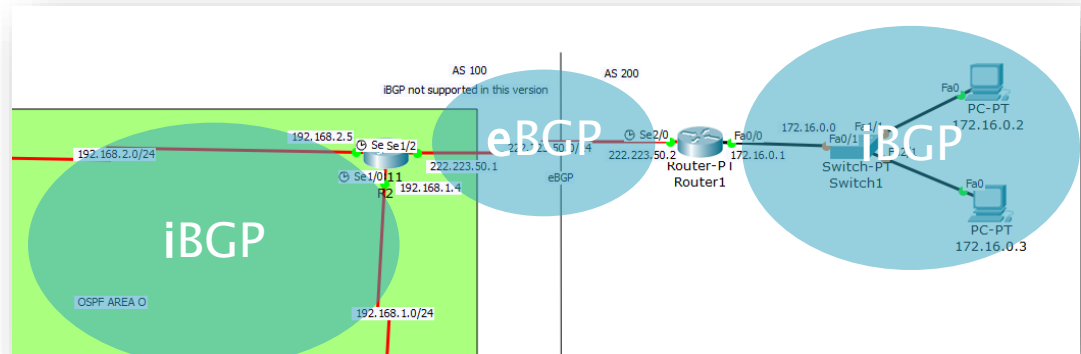
- ▶ Interior Gateway Protocols (IGP):
 - Used by Gateways to exchange routing information within (inside) an Autonomous System (AS)
 - E.g., corporate network, campus network, ISP network, ...
 - Operation: Distance vector (RIP, IGRP), Link state (OSPF, IS-IS)
- ▶ Exterior Gateway Protocol (EGP):
 - Used by Gateways to exchange routing information between (outside) Autonomous Systems
 - Routing path selection is based on:
 - Network policies
 - Network administrator configured rule-sets
 - Operation: Path vector (BGP)

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Introduction

- ▶ Internet uses BGP4 (Border Gateway Protocol version 4):
 - Was published in 2006 as RFC 4271
 - eBGP (external BGP): exchange routing information b/w Autonomous Systems
 - iBGP (internal BGP): advertise routes within the same Autonomous System
- ▶ AS neighbors:
 - BGP Advertisement
 - IP/Prefix (reachability)
 - Update routing Paths



BGP and Internet

- ▶ BGP uses Path vector

```
// Received at 15:32:43 (2.6 second delay)
{
  "timestamp": 1682688760.45,
  "peer": "193.203.0.75",
  "peer_asn": "40994",
  "id": "193.203.0.75-0187c810da820004",
  "host": "rrc05.ripe.net",
  "type": "UPDATE",
  "path": [40994, 174, 37288, 37288, 37288, 37288, 37288, 37288, 327992, 328184],
  "community": [[174, 21101], [174, 22012], [17152, 1]],
  "origin": "IGP",
  "announcements": [
    {
      "next_hop": "193.203.0.75",
      "prefixes": [
        "155.12.224.0/19"
      ]
    }
  ],
  "withdrawals": []
}
```

*Screenshot from: RIPE NCC, RIS Live.
Retrieved from <https://ris-live.ripe.net/>*

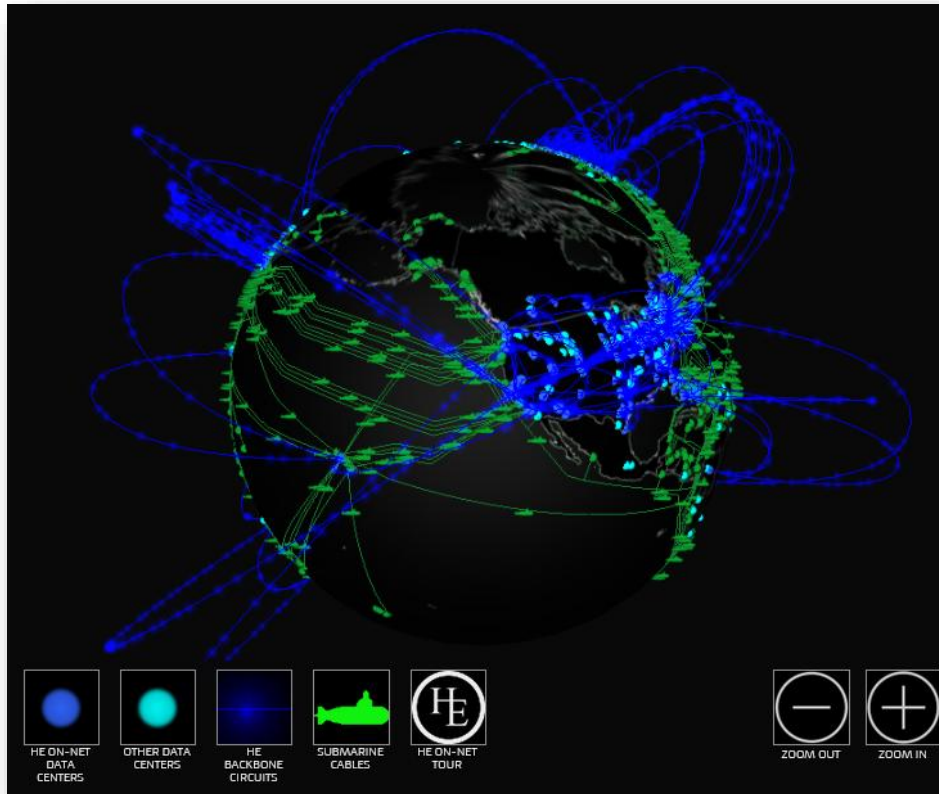
A screenshot of a web browser showing the 'Index of /bgpdata' directory. The browser's address bar shows the URL 'https://archive.routeviews.org/bgpdata/'. The page title is 'Index of /bgpdata'. Below the title is a table with four columns: 'Name', 'Last modified', 'Size', and 'Description'. The table lists various directories and files, including 'Parent Directory', '2001.10/', '2001.11/', '2001.12/', '2002.01/', '2002.02/', '2002.03/', '2002.04/', '2002.05/', and '2002.06/'. Each entry shows the last modified date and time, and a dash in the 'Size' and 'Description' columns.

Name	Last modified	Size	Description
Parent Directory	-	-	-
2001.10/	2004-02-23 20:19	-	-
2001.11/	2004-02-23 20:20	-	-
2001.12/	2004-02-23 20:27	-	-
2002.01/	2004-02-23 20:36	-	-
2002.02/	2004-02-23 20:46	-	-
2002.03/	2004-02-23 20:56	-	-
2002.04/	2004-02-23 21:08	-	-
2002.05/	2004-02-23 21:19	-	-
2002.06/	2004-02-23 21:29	-	-

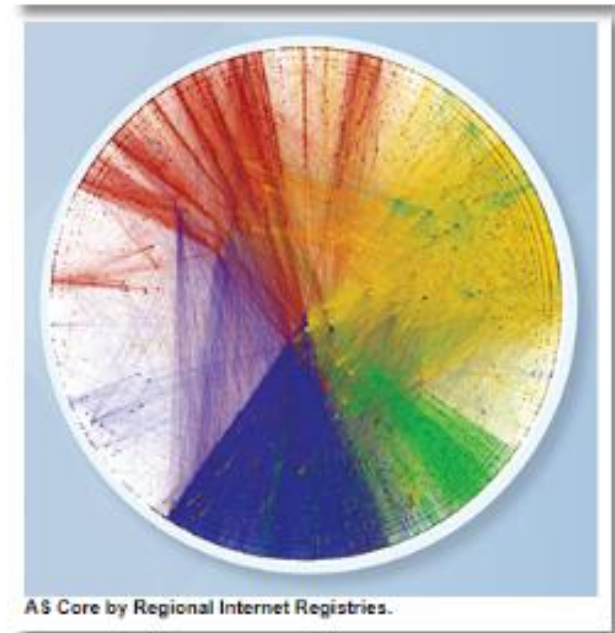
*Screenshot from: University of Oregon
Route Views Project, BGP Data Archive.
Retrieved from
<https://archive.routeviews.org/bgpdata/>*

- ▶ Routing information includes the 'path' of AS's that are used to reach the destination
- ▶ Routing Table: Each entry contains the destination network, the next router and the path to reach the destination

BGP visualizations



Screenshot from: Hurricane Electric, 3D Network Map. Retrieved from <https://www.he.net/3d-map>



Screenshots from: CAIDA, AS Core Network Visualization (2017). Retrieved from https://www.caida.org/research/topology/as_core_network/2017/

BGP security concerns

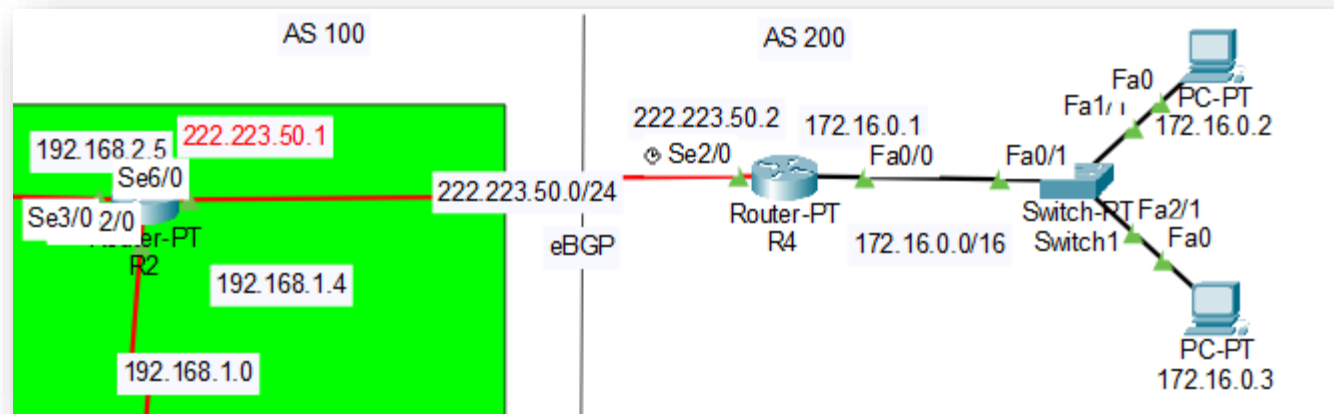
- ▶ BGP advertisement messages (often unencrypted) are generally accepted by any other router without verifying its origin or its integrity
 - **BGP route manipulation:** A malicious device alters content of the BGP table, preventing traffic from reaching to its intended destination
 - **BGP route hijacking:** A rogue device maliciously announces victim's prefixes to reroute traffic to or through itself
 - **BGP denial-of-service (DoS):** A malicious device sends unexpected or undesirable BGP traffic to a victim
 - ...
- ▶ What can be done?
 - Secure Inter-Domain Routing (SIDR)
 - BGPsec
 - ...

Exercise 11: Practical work

► Use your existing cisco packet tracer file (from last class exercise):

1. Include all your existing topology in AS 100
2. Create another AS 200
3. Connect and configure AS 100 and AS 200 using BGP:
 1. Conf t
 2. router bgp 100 //AS number
 3. Network X.X.X.X mask Z.Z.Z.Z //Subnet mask, networks on ASBR that must be advertised
 4. neighbor X.X.X.X remote-as 200 //interface IP of the other AS ASBR interface
 5. ... //all neighbors internal or external that would need BGP (note: iBGP is not supported in CPT)
 6. Exit
4. Verify using PING and TRACERT/TRACEROUTE
5. Study the BGP path selection process:
<https://www.cisco.com/c/en/us/support/docs/ip/border-gateway-protocol-bgp/13753-25.html#anc2>

Deadline: See 'Teams' Assignment section



Save cisco packet tracer file with your 'First_Last-name_Ex-3-11

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Router ACLs (operations)

- ▶ Access Control List (ACL):
 - ▶ List of Conditions → Actions: Permit/Deny
 - ▶ To control traffic flow
- ▶ Usually, network traffic goes through conditions from top to down until a match is found
 - ▶ Every ACL has an implicit DENY at the end!
- ▶ Two main types of ACLs:
 - ▶ **Standard**: Filter traffic based on **source** IP address
 - ▶ **Extended**: Filter traffic based on **both** source and destination IP addresses, **as well as other criteria** (e.g., protocol and port number)

Router ACLs (application)

- ▶ Depending on the direction of traffic (source and destination) that needs to be controlled:
 - ▶ You can apply an ACL on either an inbound or outbound interface (based on 'where you stand')
 - ▶ The 'in' direction filters traffic as it enters the interface, 'out' direction filters traffic as it leaves the interface
- ▶ ACLs are applied to traffic in a specific order, based on their number or name:
 - ▶ If Multiple entries in an ACL applied to an interface, traffic will be processed by the ACL with the lowest number or alphabetical name first, and then by the next lowest number or alphabetical name, and so on

Cisco router ACLs (syntax)

- ▶ **Standard access-list (classify traffic based on source IP): IDs 1–99 or 1300–1999, Names**

i Apply close to destination

Configure (modern syntax)

```
Rx(config)# ip access-list standard {ACL_No./Name}  
Rx(config-std-nacl)# {permit/deny} X.X.X.X Z.Z.Z.Z //X: host ip, Z:  
wildcard
```

Apply, optional: sequence no. behind permit/deny to change filter order

```
Rx(config)# interface z/z
```

```
Rx(config-if)#ip access-group {ACL_No./Name} in/out
```

- ▶ **Extended access lists (classify traffic based on source IP, destination IP, Protocol, Port no.): IDs 100–199 or 2000–2699, Names**

i Apply close to Source

Configure (modern syntax)

```
Rx(config)# ip access-list extended {ACL_No./Name}  
Rx(config-ext-nacl)# {permit/deny} <Protocol> <Source_IP> <  
Source_Wildcard> [Protocol: port no.] <Destination_IP>  
<Destination_Wildcard> [Protocol: port no.]
```

Apply, optional: sequence no. behind permit/deny to change filter order

```
Rx(config)# interface z/z
```

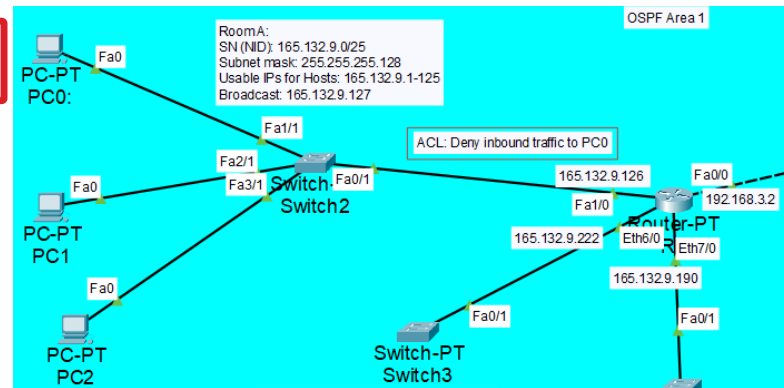
```
Rx(config-if)#ip access-group {ACL_No./Name} in/out
```

i Using aforementioned syntax instead of the classic 'ip access-group' syntax lets you define order easily

Exercise 12: Practical work

- ▶ Use your existing cisco packet tracer file (from last class exercise):
 - On R0, setup standard ACL such that: See red box:

ACL: Deny all inbound traffic to PC0



Router should:

The device sends back an ICMP Administratively Prohibited Unreachable message.

- ▶ Commands (classic syntax):
 - *R0(config)#access-list* Number? deny ip Source-IP? Destination-IP?
 - *R0(config)#int* Fa0/0
 - *R0(config-if)#ip access-group* Number? in/out?

Save cisco packet tracer file with your 'First_Last-name_Ex-3-12

Deadline: See 'Teams' Assignment section

Lecture 5 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**